

Some, however, believe that although we cannot describe it coherently, all of us have a sense of what consciousness is... and then they proceed to call that sense itself... consciousness.

Now, all of you reading this, have immediately started thinking about what consciousness is, and all of you are right (or wrong!). Because who is to tell a conscious mind what its own consciousness is? This is why people believe that it's not a scientific question, and perhaps a scientific answer can never be found. This was widely considered to be a valid proposition by most people (including scientists) until modern advancements gave us the confidence to be able to question that assertion.

TECH TO THE RESCUE!

We believe that one invention of modern medicine in particular has changed our perception of what consciousness is. MRI (Magnetic Resonance Imaging) is the technology that changed the way we look at everything in a human body, but it especially allowed us to look inside the brains of living human beings – something we couldn't ever do before.

Of course, CT (Computed Tomography) scans were available a few years before MRI, but they use X-rays, which as we all know, are best taken in as small doses as possible, or if possible, best avoided. MRI on the other hand uses very strong magnetic fields and radio waves to generate images of

inside our bodies. The biggest drawback of MRI scans is that the patient is subjected to very loud sounds, and because of the extremely strong magnetic fields that are generated, patients cannot have any sort of metal implant or foreign objects in their bodies. The MRI won its inventor, Paul C. Lauterbur, the Nobel prize in medicine in 2003.

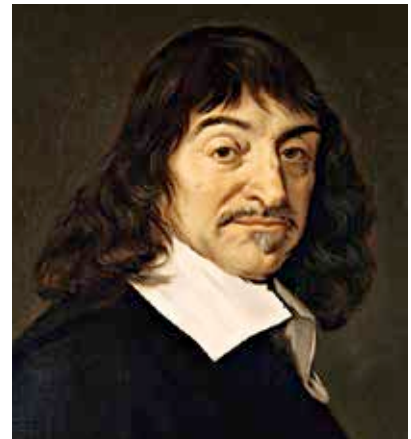
Because of the MRI, (and also CT scans to be fair) we were able to understand living brains like never before. Previous brain studies were always conducted on cadavers, because cutting open the brain of a healthy individual usually meant a very dead individual!

However, doctors were able to study every aspect of the brain in great detail, and, for example, were able to find how different areas of the brain adversely affected (by stroke, or brain damage) a normal human brain. Soon patterns started forming, and we were able to understand what each part of the brain did. We'd be lying if we said that there were no gaps in the knowledge of science today when it comes to the human brain, because there are plenty. What we are, however, is far, far less ignorant than we were before.

DISCIPLINES

Since science has been divided into various disciplines, each dealing with its own specific area of study, you'd think the study of consciousness was a purely biology-based question, and it should be biologists who attempt to

answer it. You would be wrong. Physicists, as usual, love to show off their expertise in everything, and seem to want to chime in with their own theories. Of course, their theories turn out to not be bad at all, and it's the reason why we're breaking this story into explanations based in physics, and then the other explanations of consciousness that seem promising. Why "the rest" and physics separated? Be-



It was René Descartes who famously said, "I think, therefore I am".

cause physicists are special enough to warrant their own group, and it's just easier if we separate them from the others in order to keep things peaceful... (we're joking, calm down!). We'll get to the physics in a bit... First we'll look at biological ways in which to try and define consciousness.

Obviously, the easiest way to describe consciousness is to consider it to be the ability to sense. All of our senses (taste, touch, smell, see, hear) combine to give us a sense of the world around us. Exterior to us, a sense of existing, for example.

Another way would be to look at it as something that allows us to think and react to the world around us. However, as scientific philosophers such as René Descartes advocated, there was a difference between the physical and the mental. Modern science isn't sure about this divide anymore...

POOF GONE!

Where do you go when you sleep? Where were you before you were born? When we say "you" we mean your consciousness, obviously. And it's a valid question. Every day, all of us lose consciousness when we sleep, often a dreamless sleep. Some of the unlucky few amongst us might need to be given general anaesthesia in order to have surgery performed on us... where do we go when under anaesthesia? An even smaller portion of us would have had a near death experience, and undoubtedly, would have "died" before being brought back to life. Where were we when we were dead?



An MRI machine helps us look inside our own brains